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Roll No

EC-4004-CBGS

B.E. IV Semester

Examination, December 2020

Choice Based Grading System (CBGS)

Communication Systems

Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.
iii) Assume any missing data.

1. a) Find the Fourier transform of exponential pulse.
b) State and prove scaling and time Convolution property of Fourier transform.
2. Derive the expression of amplitude modulated signal when a carrier $E_c \cos \omega_c t$ is modulated by $E_m \cos \omega_m t$. Also derive the expression of efficiency.
3. a) Give a comparison between Amplitude Modulation and Frequency Modulation.
b) An AM broadcast transmitter radiates 20 kW when the modulation index is 75%. Calculate
i) How much of this carrier power.
ii) Power of each sidebands.
4. a) What is the need for modulation in communication system?
b) Explain synchronous detection of DSB-SC signal with a neat diagram.

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5.
 - a) Explain the types of angle modulation and establish a relation between them.
 - b) A carrier $A \cos \omega_c t$ is frequency modulated by a single tone modulating signal $E_m \cos \omega_m t$. Find an expression for the FM wave.
6.
 - a) Draw and explain the block diagram of producing PM from FM. What is the necessity of using frequency multipliers with PM.
 - b) Explain pre-emphasis and de-emphasis.
7.
 - a) Draw and explain the block diagram of FM receiver.
 - b) Define AGC? Explain a simple AGC circuit diagram.
8. Write short notes on any two of the following:
 - a) Correlation (auto and cross)
 - b) Errors in synchronously detection
 - c) Image signal rejection, selectivity, sensitivity and fidelity
 - d) FM generation
 - e) FM transmitter and receiver
